

Part A — User and System Investigation (10 minutes)

Determine the username of the account you are currently using.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ whoami
ubuntu
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Display your UID, primary GID, and supplementary groups.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ id
uid=1000(ubuntu) gid=1000(ubuntu) groups=1000(ubuntu),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),111(lpadmin),114(lxd)
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Display all users currently logged in to the system.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ w
 09:44:32 up 4 min,  1 user,  load average: 0.24, 0.31, 0.16
USER          TTY          FROM          LOGIN@   IDLE   JCPU   PCPU   WHAT
ubuntu        tty2          -              09:40    4:13   0.01s  0.01s  /usr/libexec/gnome-session-init-worker ubuntu
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Determine what activities the currently logged-in users are performing.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps -ef
UID          PID    PPID  C  TIME TTY          TIME CMD
root           1      0  0  09:40 ?        00:00:01 /usr/lib/systemd/systemd --switched-root --system --deserialize=51 auto automatic-ubiquity noprompt
root           2      0  0  09:40 ?        00:00:00 [kthreadd]
root           3      2  0  09:40 ?        00:00:00 [pool_workqueue_release]
root           4      2  0  09:40 ?        00:00:00 [kworker/R-rcu_gp]
root           5      2  0  09:40 ?        00:00:00 [kworker/R-sync_wq]
root           6      2  0  09:40 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7      2  0  09:40 ?        00:00:00 [kworker/R-slub_flushwq]
root           8      2  0  09:40 ?        00:00:00 [kworker/R-netns]
root           9      2  0  09:40 ?        00:00:00 [kworker/0:0-events]
root          10      2  0  09:40 ?        00:00:00 [kworker/0:0H-kblockd]
root          11      2  0  09:40 ?        00:00:00 [kworker/0:1-cgroup_free]
root          12      2  0  09:40 ?        00:00:00 [kworker/u512:0-ipv6_addrconf]
root          13      2  0  09:40 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14      2  0  09:40 ?        00:00:00 [ksoftirqd/0]
root          15      2  0  09:40 ?        00:00:00 [rcu_preempt]
root          16      2  0  09:40 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/1]
root          17      2  0  09:40 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          18      2  0  09:40 ?        00:00:00 [migration/0]
root          19      2  0  09:40 ?        00:00:00 [kprobe-optimizer]
root          20      2  0  09:40 ?        00:00:00 [idle_inject/0]
root          21      2  0  09:40 ?        00:00:00 [cpuhp/0]
root          22      2  0  09:40 ?        00:00:00 [cpuhp/1]
root          23      2  0  09:40 ?        00:00:00 [idle_inject/1]
root          24      2  0  09:40 ?        00:00:00 [migration/1]
root          25      2  0  09:40 ?        00:00:00 [ksoftirqd/1]
root          26      2  0  09:40 ?        00:00:00 [kworker/1:0-cgroup_free]
root          27      2  0  09:40 ?        00:00:00 [kworker/1:0H-kblockd]
root          28      2  0  09:40 ?        00:00:00 [kworker/u513:0-async]
root          29      2  0  09:40 ?        00:00:00 [kworker/u514:0-flush-8:0]
root          30      2  0  09:40 ?        00:00:00 [kdevtmpfs]
root          31      2  0  09:40 ?        00:00:00 [kworker/R-inet_frag_wq]
root          32      2  0  09:40 ?        00:00:00 [rcu_tasks_kthread]
root          33      2  0  09:40 ?        00:00:00 [rcu_tasks_rude_kthread]
root          34      2  0  09:40 ?        00:00:00 [kautdtd]
root          35      2  0  09:40 ?        00:00:00 [khungtaskd]
root          36      2  0  09:40 ?        00:00:00 [oom_reaper]
root          37      2  0  09:40 ?        00:00:00 [kworker/u513:1-async]
root          38      2  0  09:40 ?        00:00:00 [kworker/u513:2-async]
root          39      2  0  09:40 ?        00:00:00 [kworker/R-writeback]
root          40      2  0  09:40 ?        00:00:00 [kcompactd0]
root          41      2  0  09:40 ?        00:00:00 [ksmd]
root          42      2  0  09:40 ?        00:00:00 [khugepaged]
root          43      2  0  09:40 ?        00:00:00 [kworker/R-kblockd]
root          44      2  0  09:40 ?        00:00:00 [kworker/R-blkcg_punt_bio]
root          45      2  0  09:40 ?        00:00:00 [kworker/R-kintegrityd]
```

Examine /etc/passwd and identify the following information for your current account:

Username

UID

GID

Home directory

Login shell

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ grep "^$(whoami):" /etc/passwd
ubuntu:x:1000:1000:ubuntu:/home/ubuntu:/bin/bash
ubuntu@ubuntu-VMware-Virtual-Platform:~$ getent passwd "$(whoami)"
ubuntu:x:1000:1000:ubuntu:/home/ubuntu:/bin/bash
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Determine how many user-account entries are present in /etc/passwd.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ wc -l /etc/passwd
  Show Apps  wd
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Display only the username and UID fields from /etc/passwd.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
bin:2
sys:3
sync:4
games:5
man:6
lp:7
mail:8
news:9
uucp:10
proxy:13
www-data:33
backup:34
list:38
irc:39
_apt:42
nobody:65534
systemd-network:998
dhcpcd:996
messagebus:995
syslog:100
systemd-resolve:989
_chrony:988
tss:987
uidd:101
systemd-oom:986
whoopsie:102
dnsmasq:999
avahi:103
nm-openvpn:984
tcpdump:983
sssd:104
speech-dispatcher:105
usbmux:106
cups-pk-helper:107
fwupd-refresh:982
saned:108
geoclue:109
cups-browsed:110
pipewire:981
hplip:111
gnome-remote-desktop:980
polkitd:979
rtkit:978
colord:977
```

Find all accounts whose UID is 0. Explain the security significance of UID 0.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ awk -F: '$3 == 0 {print $1 ":" $3}' /etc/passwd
root:0
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Compare /etc/passwd and /etc/shadow. Why are password hashes stored in /etc/shadow instead of /etc/passwd?

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
_chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
tss:x:987:987:tss user for tpm2:/:/usr/sbin/nologin
uidd:x:101:105::/run/uidd:/usr/sbin/nologin
systemd-oom:x:986:986:systemd Userspace OOM Killer:/:/usr/sbin/nologin
whoopsie:x:102:108::/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:103:109:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
nm-openvpn:x:984:984:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin
tcpdump:x:983:983:tcpdump:/nonexistent:/usr/sbin/nologin
sssd:x:104:110:SSSD system user:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:105:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false
usbmux:x:106:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
cups-pk-helper:x:107:111:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:982:982:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:108:112::/var/lib/saned:/usr/sbin/nologin
geoclue:x:109:113::/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:110:111::/nonexistent:/usr/sbin/nologin
pipewire:x:981:981:system user for pipewire:/nonexistent:/usr/sbin/nologin
hplip:x:111:7:HPLIP system user:/run/hplip:/bin/false
gnome-remote-desktop:x:980:980:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
```



```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo cat /etc/shadow
[sudo: authenticate] Password:
root:*:20566:0:99999:7:::
daemon:*:20566:0:99999:7:::
bin:*:20566:0:99999:7:::
sys:*:20566:0:99999:7:::
sync:*:20566:0:99999:7:::
games:*:20566:0:99999:7:::
man:*:20566:0:99999:7:::
lp:*:20566:0:99999:7:::
mail:*:20566:0:99999:7:::
news:*:20566:0:99999:7:::
uucp:*:20566:0:99999:7:::
proxy:*:20566:0:99999:7:::
www-data:*:20566:0:99999:7:::
backup:*:20566:0:99999:7:::
list:*:20566:0:99999:7:::
irc:*:20566:0:99999:7:::
_apt:*:20566:0:99999:7:::
nobody:*:20566:0:99999:7:::
systemd-network:!:20566::::::1:
dhcpcd:!:20566::::::1:
messagebus:!:20566::::::
syslog:!:20566::::::
systemd-resolve:!:20566::::::1:
_chrony:!:20566::::::
tss:!:20566::::::
uidd:!:20566::::::
systemd-oom:!:20566::::::1:
whoopsie:!:20566::::::
dnsmasq:!:20566::::::
avahi:!:20566::::::
nm-openvpn:!:20566::::::1:
tcpdump:!:20566::::::1:
sssd:!:20566::::::
speech-dispatcher:!:20566::::::
usbmux:!:20566::::::
cups-pk-helper:!:20566::::::
fwupd-refresh:!:20566::::::
saned:!:20566::::::
geoclue:!:20566::::::
cups-browsed:!:20566::::::
nfs-wire:!:20566::::::
```

Part B — Creating and Verifying Users (20 Minutes)

Create the following users and ensure that a home directory is created for each:

alice, bob, Charlie

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo useradd -m alice
sudo useradd -m bob
sudo useradd -m charlie
```

Verify that all three accounts have been added to /etc/passwd.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ grep -E '^(alice|bob|charlie):' /etc/passwd
alice:x:1001:1002::/home/alice:/bin/sh
bob:x:1002:1003::/home/bob:/bin/sh
charlie:x:1003:1004::/home/charlie:/bin/sh
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Determine the UID assigned to each newly created user.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ id -u alice
id -u bob
id -u charlie
1001
1002
1003
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Determine the home directory and default shell assigned to alice.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ getent passwd alice
alice:x:1001:1002::/home/alice:/bin/sh
ubuntu@ubuntu-VMware-Virtual-Platform:~$ getent passwd alice | cut -d: -f6
/home/alice
ubuntu@ubuntu-VMware-Virtual-Platform:~$ getent passwd alice | cut -d: -f7
/bin/sh
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Inspect /home and verify that the required home directories were created.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ls -la /home
total 24
drwxr-xr-x  6 root    root    4096 Sep  1 09:56 .
drwxr-xr-x 20 root    root    4096 Aug 17 20:29 ..
drwxr-x---  2 alice   alice   4096 Sep  1 09:56 alice
drwxr-x---  2 bob     bob     4096 Sep  1 09:56 bob
drwxr-x---  2 charlie charlie 4096 Sep  1 09:56 charlie
drwxr-x--- 16 ubuntu  ubuntu 4096 Aug 17 22:38 ubuntu
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Set passwords for all three accounts.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo passwd alice
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo passwd bob
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo passwd charlie
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Switch from your current account to alice using a login shell. Verify the change using an appropriate command, and then return to your original account.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ su - alice
Password:
$ whoami
alice
$ id
uid=1001(alice) gid=1002(alice) groups=1002(alice)
$ pwd
/home/alice
$ exit
ubuntu@ubuntu-VMware-Virtual-Platform:~$ whoami
ubuntu
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Part C — Group Management and Access Control (30 Minutes)

Create two groups:

security and developers

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo groupadd security
sudo groupadd developers
ubuntu@ubuntu-VMware-Virtual-Platform:~$ getent group security
getent group developers
security:x:1005:
developers:x:1006:
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Add alice and charlie to the security group.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo usermod -aG security alice
sudo usermod -aG security charlie
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Add bob to the developers group.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo usermod -aG developers bob
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Without switching accounts, verify the group membership of all three users.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ id alice
id bob
id charlie
uid=1001(alice) gid=1002(alice) groups=1002(alice),1005(security)
uid=1002(bob) gid=1003(bob) groups=1003(bob),1006(developers)
uid=1003(charlie) gid=1004(charlie) groups=1004(charlie),1005(security)
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Display the entries corresponding to security and developers from /etc/group.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ grep -E '^(security|developers):' /etc/group
security:x:1005:alice,charlie
developers:x:1006:bob
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Add bob to the security group without removing him from the developers group.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo usermod -aG security bob
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```


Verify that bob now belongs to both groups.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ id bob
uid=1002(bob) gid=1003(bob) groups=1003(bob),1005(security),1006(developers)
ubuntu@ubuntu-VMware-Virtual-Platform:~$ groups bob
bob : bob security developers
ubuntu@ubuntu-VMware-Virtual-Platform:~$ grep -E '^(security|developers):' /etc/group
security:x:1005:alice,charlie,bob
developers:x:1006:bob
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```